

A transparent rule-based algorithm for tillage detection from Sentinel-1 time series

Dries [Surname][1], [Co-author][1], [Co-author][2]
[1]UCLouvain – Geomatics, Louvain-la-Neuve, Belgium [2][Affiliation] · dries.debievre@uclouvain.be

[Institution logo(s)]

1 Introduction and definitions

[Context — why monitoring tillage matters for soil protection, and what is missing in current approaches.]

[Objective — what the algorithm does and why a transparent, rule-based approach.]

[Study area and data — region, years, reference/field data.]

Definitions

Tillage event [definition]
Inversion tillage [definition, e.g. ploughing]
Coherence γ [definition]
Backscatter σ^0 [definition]
Cross-ratio $\sigma_{VH}^0/\sigma_{VV}^0$ [definition]

2 Workflow

Phase 1 · Tillage event detection

Coherence
γ_{VV}
Sentinel-1

Coherence change threshold
$\tau_{\Delta\gamma}$
tunable parameter

1a. VV coherence change detection
For each orbit, flag a detection where the change in coherence between consecutive acquisitions exceeds the threshold:

$$\gamma_{VV}^{a+1} - \gamma_{VV}^{a} > \tau_{\Delta\gamma}$$
or
$$\gamma_{VV}^{a-1} - \gamma_{VV}^{a} > \tau_{\Delta\gamma}$$

Snow-affected dates
ERA5-Land + Sentinel-1

1b. Filter detections due to snowfall
Discard detections falling on a masked date. Masked dates are those identified as snow-affected from ERA5-Land data and the ERA5 grid-cell averaged Sentinel-1 coherence time series.

NDVI
Sentinel-2

1c. Filter for vegetation presence
Discard a detection if **NDVI** > 0.25 within 10 days after it; if NDVI is missing, use CR > 0.15 within 7 days.

Cross-ratio
$\sigma_{VH}^0 / \sigma_{VV}^0$
Sentinel-1

Backscatter
σ_{VV}^0, σ_{VH}^0
Sentinel-1

Backscatter thresholds
τ_{VV}, τ_{VH}
tunable parameters

Phase 2 · Inversion tillage discrimination
Flag **VV** if $\sigma_{VV}^0 > \tau_{VV}$; flag **VH** if $\sigma_{VH}^0 > \tau_{VH}$.

Phase 3 · Multi-orbit consolidation
Merge consecutive detections from different orbits into a single event, and compute the fraction flagged as inversion tillage (VV and VH separately).

Phase 4 · Confidence indicator for inversion tillage
Assign a confidence level (High / Medium / Low) to each event, based on the fraction flagged as inversion tillage from VV and VH backscatter.

3 Performance

[Figure — e.g. example parcel time series: coherence, backscatter, NDVI with detected and reference tillage dates]

Figure 1. [Caption]

	Precision	Recall	F1-score
Tillage event detection	[ ]	[ ]	[ ]
Inversion tillage (VV)	[ ]	[ ]	[ ]
Inversion tillage (VH)	[ ]	[ ]	[ ]

Table 1. [Caption — validation against reference data]

[Figure — e.g. confidence level (High / Medium / Low) vs. reference]

Key results

[Result 1 — headline number]

[Result 2 — effect of the snow / vegetation filters]

[Result 3 — where it fails, and why]

4 Conclusions

1
[Key message — one sentence the reader should remember]

2
[Key message — one sentence the reader should remember]

3
[Key message — one sentence the reader should remember]

[QR code: paper / code / contact]

[References & acknowledgements]

2nd EO 4 Soil Symposium
ESA SYMPOSIUM ON EARTH OBSERVATION FOR SOIL PROTECTION AND RESTORATION
04–06 November 2026 | ESA–ESRIN | Frascati (Rome), Italy